

Design Basis & Open Issues Register — Rev.01

Project: Fully underground civil shelter, private plot, Rusanivski Sady, Kyiv Document status: Rev.01 — CONTROLLING DOCUMENT Prepared: 13.09.2026 Supersedes: Nothing (first issue). This register itself now governs the status of the earlier reference-concept package (PDF/XLSX/PNG/SVG) produced in this session — see RECLASSIFICATION_NOTICE.md.

1. Purpose and effect of this register

This register records:

- 1 The design basis — what is known/assumed about the project, and on what authority (client brief vs. assumption vs. confirmed survey/test data).
- 2 The open issues — the specific inputs and calculations that are currently missing, and that block any drawing, quantity, or budget figure from being treated as project data.

Until every item in Section 4 is closed (status changed from OPEN to CLOSED, with the closing document referenced), no geometry, reinforcement quantity, concrete quantity, budget figure, or KZh-numbered drawing may be issued, quoted from, procured against, or built from. This applies retroactively to the reference-concept package generated earlier in this session.

Only a licensed structural engineer (or a geotechnical/survey professional for their respective disciplines) may close an item or issue a superseding calculation, drawing, or quantity.

2. Design basis (as stated by the client — not yet confirmed by survey or test)

#	Item	Value stated by client	Status
1	Location	Kyiv, Rusanivski Sady, private plot	Client statement — not surveyed
2	Occupancy	4 persons	Client statement
3	Internal clear dimensions	4000 × 3000 mm, clear height 2200 mm	Client-proposed, not structurally verified
4	Structural concept	Monolithic reinforced-concrete box	Client-proposed concept
5	Waterproofing concept	External membrane not feasible → "white box" (waterproof-concrete) concept	Client-proposed concept, not confirmed by a waterproofing/structural engineer
6	Concrete class (indicative only)	C35/45, W12, F200	Client-proposed starting point, not a design value
7	Reinforcement class (indicative only)	A500C	Client-proposed starting point, not a design value
8	Entrance opening (indicative only)	900 mm	Client-proposed, pending door supplier data (see Doc.04)
9	Emergency hatch (indicative only)	800 × 800 mm	Client-proposed, pending supplier data (see Doc.04)

All values in this section are inputs to be tested, not outputs to be relied on. None of them, alone or combined, constitute a structural design.

3. Owner / responsibility matrix (roles only — names assigned once contractors/professionals are selected)

Six roles are used throughout this register and the rest of the handover package. No individual names are recorded until each role is actually filled; this matrix and the Owner column in Section 4 are updated with names at that point, not before.

Role	Definition for this project
Client	Property owner / party commissioning the shelter
Structural Engineer	Licensed structural engineer who will assign the protection class, run calculations, and issue IFC drawings
Geotechnical Engineer	Licensed professional performing the geotechnical investigation (Doc.03)
Surveyor	Licensed professional performing the site/topographic survey (Doc.05)
Supplier	Door, emergency-hatch, ventilation, and concrete suppliers responding to Doc.04
Contractor	Building contractor — not yet engaged; reserved for the construction phase once IFC drawings exist

Document	Primary owner (role)	Also involves
01 — This register	Structural Engineer	Client
02 — Engineer Handover Sheet	Client	Structural Engineer
03 — Geotechnical Investigation Brief	Client	Geotechnical Engineer
04 — Supplier Data Request Package	Client	Supplier
05 — Site/Topographic Survey Brief	Client	Surveyor
Reference KZh package (frozen, superseded)	— none —	not applicable
Future: IFC drawings & calculation package	Structural Engineer	Contractor (recipient, once issued)

4. Open issues register (BLOCKING)

#	Open issue	Why it blocks the design	Owner (role)	Status	Closing document
OI-01	Geotechnical investigation not performed (soil profile, groundwater level, unit weight, friction angle, bearing capacity, aggressivity)	Governs bearing design, lateral earth pressure, uplift/UPL check, and waterproofing detailing	Client, Geotechnical Engineer	OPEN	Geotechnical investigation report (commissioned via Doc.03)

#	Open issue	Why it blocks the design	Owner (role)	Status	Closing document
OI-02	Site / topographic survey not performed (existing ground level, boundaries, neighboring foundations and utilities, access constraints)	Governs excavation depth, setbacks, and buildability	Client, Surveyor	OPEN	Topographic survey report (commissioned via Doc.05)
OI-03	Design/surcharge loading not defined (vehicle access, snow, future landscaping/canopy over the structure)	Required input to slab and wall design loads	Structural Engineer	OPEN	Loading brief from licensed engineer
OI-04	Protection class of the civil shelter not assigned	Governs wall/slab thickness, door class, ventilation filtration requirements, and applicable code clauses	Structural Engineer	OPEN	Protection-class determination memo
OI-05	Structural calculations not performed (walls, slabs, uplift/UPL check, lateral earth pressure, opening reinforcement)	This is the core structural design — nothing here has been calculated to a verifiable standard	Structural Engineer	OPEN	Calculation package + issued-for-construction (IFC) drawings
OI-06	Door, emergency-hatch, and ventilation equipment not selected	Structural openings, embedded parts, and penetrations cannot be finalized without supplier dimensional/technical data	Client, Supplier	OPEN	Supplier datasheets (requested via Doc.04)
OI-07	Concrete mix design and supplier not confirmed	W/F class, admixtures (for the "white box" concept), and pumpability depend on an actual mix design from a plant	Client, Supplier	OPEN	Concrete mix design + supplier confirmation (requested via Doc.04)

#	Open issue	Why it blocks the design	Owner (role)	Status	Closing document
OI-08	Applicable current code edition not confirmed (ДБН/ДСТУ and civil-defense shelter requirements)	Design must cite the code editions in force at the time of the calculation, confirmed by the licensed engineer	Structural Engineer	OPEN	Reference list in calculation package

Rule: an item may only move to CLOSED when the named closing document exists and is referenced here by date and author. No item may be closed by inference, by re-running the earlier reference-concept calculations, or by this or any other AI-generated output.

5. Revision history

Rev	Date	Change	Author
01	13.09.2026	First issue. Freezes prior reference-concept package (geometry, quantities, budgets, KZh-0...KZh-17) pending closure of OI-01...OI-08. Owner/responsibility matrix added (roles only).	Assembled per client instruction

6. Explicit exclusions (per client instruction, Rev.01)

The following are out of scope for this register and for every other document in the handover package, and must not be produced until OI-01...OI-08 are closed by the responsible professionals:

- New or revised reinforcement quantities, bar schedules, or diameters
- New or revised concrete volumes or mix quantities
- New or revised budget/estimate figures
- New, revised, or re-issued KZh-numbered drawings
- Uplift (UPL), lateral earth pressure, slab, or reinforcement conclusions